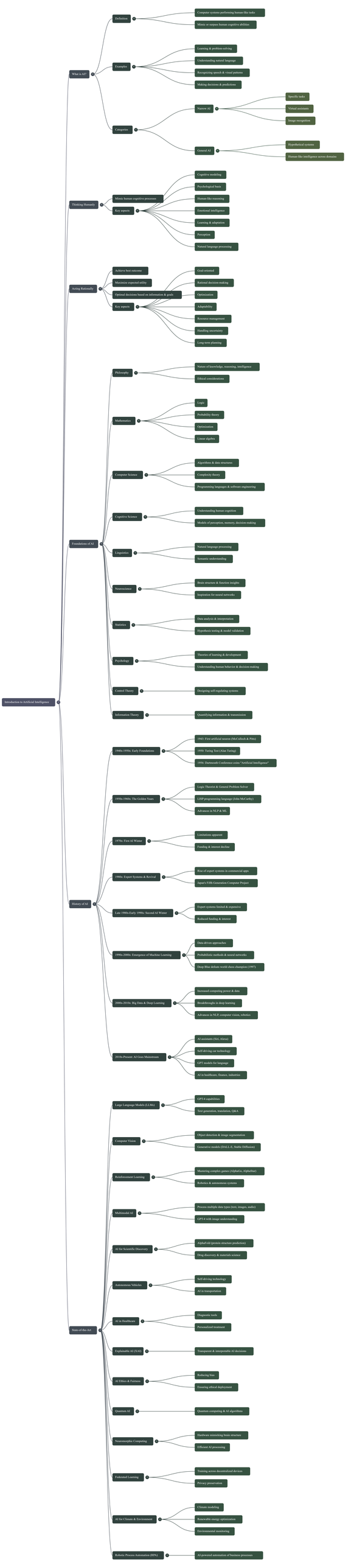
Introduction to Artificial Intelligence
What is AI?
Definition
Computer systems performing human-like tasks
Mimic or surpass human cognitive abilities
Examples
Learning & problem-solving
Understanding natural language
Recognizing speech & visual patterns
Making decisions & predictions
Categories
Narrow AI
Specific tasks
Virtual assistants
Image recognition
General AI
Hypothetical systems
Human-like intelligence across domains
Thinking Humanly
Mimic human cognitive processes
Key aspects
Cognitive modeling
Psychological basis
Human-like reasoning
Emotional intelligence
Learning & adaptation
Perception
Natural language processing
Acting Rationally
Achieve best outcome
Maximize expected utility
Optimal decisions based on information & goals
Key aspects
Goal-oriented
Rational decision-making
Optimization
Adaptability
Resource management
Handling uncertainty
Long-term planning
Foundations of AI
Philosophy
Nature of knowledge, reasoning, intelligence
Ethical considerations
Mathematics
Logic
Probability theory
Optimization
Linear algebra
Computer Science
Algorithms & data structures
Complexity theory
Programming languages & software engineering
Cognitive Science
Understanding human cognition
Models of perception, memory, decision-making
Linguistics
Natural language processing
Semantic understanding
Neuroscience
Brain structure & function insights
Inspiration for neural networks
Statistics
Data analysis & interpretation
Hypothesis testing & model validation
Psychology
Theories of learning & development
Understanding human behavior & decision-making
Control Theory
Designing self-regulating systems
Information Theory
Quantifying information & transmission
History of AI
1940s-1950s: Early Foundations
1943: First artificial neuron (McCulloch & Pitts)
1950: Turing Test (Alan Turing)
1956: Dartmouth Conference coins "Artificial Intelligence"
1950s-1960s: The Golden Years
Logic Theorist & General Problem Solver
LISP programming language (John McCarthy)
Advances in NLP & ML
1970s: First AI Winter
Limitations apparent
Funding & interest decline
1980s: Expert Systems & Revival
Rise of expert systems in commercial apps
Japan's Fifth Generation Computer Project
Late 1980s-Early 1990s: Second AI Winter
Expert systems limited & expensive
Reduced funding & interest
1990s-2000s: Emergence of Machine Learning
Data-driven approaches
Probabilistic methods & neural networks
Deep Blue defeats world chess champion (1997)
2000s-2010s: Big Data & Deep Learning
Increased computing power & data
Breakthroughs in deep learning
Advances in NLP, computer vision, robotics
2010s-Present: AI Goes Mainstream
AI assistants (Siri, Alexa)
Self-driving car technology
GPT models for language
AI in healthcare, finance, industries
State-of-the-Art
Large Language Models (LLMs)
GPT-4 capabilities
Text generation, translation, Q&A
Computer Vision
Object detection & image segmentation
Generative models (DALL-E, Stable Diffusion)
Reinforcement Learning
Mastering complex games (AlphaGo, AlphaStar)
Robotics & autonomous systems
Multimodal AI
Process multiple data types (text, images, audio)
GPT-4 with image understanding
AI for Scientific Discovery
AlphaFold (protein structure prediction)
Drug discovery & materials science
Autonomous Vehicles
Self-driving technology
AI in transportation
AI in Healthcare
Diagnostic tools
Personalized treatment
Explainable AI (XAI)
Transparent & interpretable AI decisions
AI Ethics & Fairness
Reducing bias
Ensuring ethical deployment
Quantum AI
Quantum computing & AI algorithms
Neuromorphic Computing
Hardware mimicking brain structure
Efficient AI processing
Federated Learning
Training across decentralized devices
Privacy preservation
AI for Climate & Environment
Climate modeling
Renewable energy optimization
Environmental monitoring
Robotic Process Automation (RPA)
AI-powered automation of business processes